

```
Open  ▾  +  *q1.c  Save  ≡  x
/mnt/c/Users/Dhanush/OneDrive/Desktop/OS/2520030014_Practical

1 #include <stdio.h>
2 #include <unistd.h>
3 #include <sys/wait.h>
4
5 int main()
6 {
7     char cmd[100];
8
9     printf("Enter Linux command: ");
10    scanf("%s", cmd);
11
12    int pid = fork();
13
14    if (pid == 0)
15    {
16        printf("Child PID: %d\n", getpid());
17        execlp(cmd, cmd, NULL);
18    }
19    else
20    {
21        printf("Parent PID: %d\n", getpid());
22        wait(NULL);
23        printf("Child completed\n");
24    }
25
26    return 0;
27 }
```

Bracket match found on line: 6

C ▾ Tab Width: 8 ▾ Ln 27, Col 2 INS

```
dhanush@LAPTOP-GF5CTKJB x  +  ▾
dhanush@LAPTOP-GF5CTKJB:/mnt/c/Users/Dhanush/OneDrive/Desktop/OS/2520030014_Practical$ gedit q1.c

** (gedit:953): WARNING **: 07:39:37.369: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:953): WARNING **: 07:39:37.369: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
dhanush@LAPTOP-GF5CTKJB:/mnt/c/Users/Dhanush/OneDrive/Desktop/OS/2520030014_Practical$ gcc q1.c -o q1
dhanush@LAPTOP-GF5CTKJB:/mnt/c/Users/Dhanush/OneDrive/Desktop/OS/2520030014_Practical$ ./q1
Enter Linux command: ls
Parent PID: 1887
Child PID: 1888
q1 q1.c
Child completed
```

```

dhanush@LAPTOP-GF5CTKJB: /mnt/c/Users/Dhanush/OneDrive/Desktop/OS/2520030014_Practical$ uname -a
Linux LAPTOP-GF5CTKJB 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
dhanush@LAPTOP-GF5CTKJB: /mnt/c/Users/Dhanush/OneDrive/Desktop/OS/2520030014_Practical$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:         48 bits physical, 48 bits virtual
Byte Order:            Little Endian
CPU(s):                16
On-line CPU(s) list:   0-15
Vendor ID:             AuthenticAMD
Model name:            AMD Ryzen 7 5825U with Radeon Graphics
CPU family:            25
Model:                 80
Thread(s) per core:    2
Core(s) per socket:    8
Socket(s):             1
Stepping:              0
BogoMIPS:              3992.39
Flags:                 fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat
                      pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pd
                      pelg b rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable
                      nonstop_tsc cpuid extd_apicid tsc_known_freq pni pclmulqdq ssse3
                      fma cx16 sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hy
                      pervisor lahf_lm cmp_legacy svm cr8_legacy abm sse4a misalignsse

```

```

all fsgsbase bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves clzero xsaveerptr arat npt nrp_save tsc_scale vmcb_clean flushbyasid decodeassists pausefilter pfthreshold v_vmsave_vmload umip vaes vpcplmulqdq rdpid fsrm

Virtualization features:
  Virtualization:      AMD-V
  Hypervisor vendor:   Microsoft
  Virtualization type: full
Caches (sum of all):
  L1d:                 256 KiB (8 instances)
  L1i:                 256 KiB (8 instances)
  L2:                  4 MiB (8 instances)
  L3:                  16 MiB (1 instance)
NUMA:
  NUMA node(s):        1
  NUMA node0 CPU(s):   0-15
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite:           Not affected
  Indirect target selection: Not affected
  Itlb multihit:        Not affected
  L1tf:                 Not affected
  Mds:                  Not affected
  Meltdown:             Not affected
  Mmio stale data:      Not affected

```

```
dhanush@LAPTOP-GF5CTKJB: ~$ cat /sys/devices/virtual/misc/meltdown
Meltdown: Not affected
Mmio stale data: Not affected
Old microcode: Not affected
Reg file data sampling: Not affected
Retbleed: Not affected
Spec rstack overflow: Vulnerable: Safe RET, no microcode
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation; Retpolines; IBPB conditional; IBRS_FW; STIBP always-on; RSB filling; PBRSE-eIBRS Not affected; BHI Not affected
Srbds: Not affected
Tsa: Vulnerable: No microcode
Tsx async abort: Not affected
Vmscape: Not affected
dhanush@LAPTOP-GF5CTKJB: /mnt/c/Users/Dhanush/OneDrive/Desktop/OS/2520030014_Practical$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 356.9M 1 disk
sdb 8:16 0 159.4M 1 disk
sdc 8:32 0 2G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk /mnt/wslg/distro
/
dhanush@LAPTOP-GF5CTKJB: /mnt/c/Users/Dhanush/OneDrive/Desktop/OS/2520030014_Practical$ ps
PID TTY TIME CMD
329 pts/0 00:00:00 bash
5495 pts/0 00:00:00 ps
```

```
dhanush@LAPTOP-GF5CTKJB: /mnt/c/Users/Dhanush/OneDrive/Desktop/OS/2520030014_Practical$ top
top - 07:47:19 up 11 min, 1 user, load average: 0.00, 0.05, 0.01
Tasks: 34 total, 1 running, 33 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 6849.7 total, 6123.9 free, 624.8 used, 251.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 6224.8 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24596   15464  11500 S   0.0   0.2   0:01.31 systemd
    2 root        20   0    3180    2208   2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    7 root        20   0    3180    2040   1940 S   0.0   0.0   0:00.00 init
   50 root       19  -1   50416   16824  15520 S   0.0   0.2   0:00.36 systemd-journal
   82 systemd+   20   0   22540   14576  11956 S   0.0   0.2   0:00.17 systemd-resolve
   89 root       20   0   35248   12252   9120 S   0.0   0.2   0:00.93 systemd-udev
  134 root       20   0    2888    1952   1820 S   0.0   0.0   0:00.11 chronyd-starter
  138 root       20   0    4472    3076   2828 S   0.0   0.0   0:00.02 cron
  141 message+   20   0    8796   5348   4652 S   0.0   0.1   0:00.10 dbus-daemon
  170 root       20   0   44092  29560  14416 S   0.0   0.4   0:00.43 networkd-dispat
  175 root       20   0   18436   9352   8108 S   0.0   0.1   0:00.21 systemd-logind
  190 syslog    20   0   220548   5600   4516 S   0.0   0.1   0:00.14 rsyslogd
  224 _chrony    20   0   24252   11632   7056 S   0.0   0.2   0:00.09 chronyd
  241 _chrony    20   0   12092    2448   1804 S   0.0   0.0   0:00.01 chronyd
  254 root       20   0    5492    2736   2556 S   0.0   0.0   0:00.01 agetty
  261 root       20   0   123456  32692  18076 S   0.0   0.5   0:00.37 unattended-upgr
  323 root       20   0    3188    1116    980 S   0.0   0.0   0:00.00 SessionLeader
  325 root       20   0    3204    1252   1108 S   0.0   0.0   0:00.03 Relay(329)
  329 dhanush    20   0    6268    5604   3928 S   0.0   0.1   0:00.18 bash
dhanush@LAPTOP-GF5CTKJB: /mnt/c/Users/Dhanush/OneDrive/Desktop/OS/2520030014_Practical$
```